Risk Estimation and Control in Dynamic Systems

Notes by Miguel Bueno

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1 Abstract

We consider a population of entities $\mathcal{X}$ evolving over time t, where each entity x_i has a latent true label Y_i and engagement weight E_{it}. The target estimand is the engagement-weighted prevalence $\mu_E = \mathbb{E}_{x\sim p_t}[Y(x)] \mid p_t(x) \propto E_{it}$, and the global objective is to minimize an upper confidence bound on a policy-dependent loss functional $\mathcal{L}(\tau, \mathcal{G})$ at $t+1$ by jointly optimizing the deployment policy τ and the labeling policy $\mathcal{G}$.

We have available a set of resources $\mathcal{R}$ categorized by type $\kappa = \rho(r)$.[1]. Each resource is characterized by a Resource Profile $\Phi_\kappa = \langle c_\kappa, \Omega_\kappa, \mathbf{N}_\kappa \rangle$, representing its marginal cost, processing capacity, and noise. [2] [3] Under a fixed global budget B_t and total system bandwidth Ω_t, the policies τ and $\mathcal{G}$ are constrained such that aggregate cost and throughput satisfy budget and capacity constraints (e.g., $\sum_\kappa c_\kappa n_\kappa \leq B_t$ and $\sum_\kappa \Omega_\kappa \leq \Omega_t$).

A resource type κ is said to be high-fidelity if it exhibits low noise, $\mathbf{N}_\kappa \approx \mathbf{I}$, but is constrained in cost and/or capacity, such that c_κ is large and $\Omega_\kappa \ll |\mathcal{X}|$. Conversely, a resource type κ is low-fidelity if it is inexpensive and scalable, with $c_\kappa \ll c_{\text{HF}}$ and $\Omega_\kappa \gtrsim |\mathcal{X}|$, but exhibits substantially higher noise [4]. A resource type κ is mid-fidelity if it lies between these extremes in noise, cost, and capacity, while still remaining meaningfully capacity-constrained relative to the low-fidelity resource.

High- and mid- fidelity resources are used to produce reference labels Y_i^* for estimating μ_E. We construct a sampled dataset $\mathcal{D}_{n,t} = \{X_i \mid R_{it} = 1\} \subset \mathcal{X}$ with inclusion probabilities π_{it}. To efficiently estimate μ_E, samples are drawn under an importance distribution $q(x)$ (potentially stratified over $\mathcal{H}$ and pooled over horizon T), with importance weights $w_i = p_{it} \cdot (q_{it})^{-1}$. Accounting for an availability probability $P(A_i = 1|Y_i = 1) \leq 1$ at the labeling stage, each available sampled entity X_i is evaluated by a panel of mid-fidelity resources $\mathcal{P}_i \subset \mathcal{R}$ under policy $\mathcal{G}$. Each resource $r \in \mathcal{P}_i$ produces a valuation v_{ir} [5], which is aggregated via $\mathcal{A}(\cdot)$ to produce an aggregated valuation $\hat{V}_i = \mathcal{A}(\{v_{ir}\}_{r\in\mathcal{P}_i})$ for all $x_i \in \mathcal{D}_{n,t}$. These valuations are mapped to discrete outcomes using a decision threshold η. The predicted label $\hat{Y}_i = \mathbb{I}(\hat{V}_i > \eta)$ is then used to estimate μ_E.

To minimize redundant expenditure, we implement adaptive task assignment where panel size $|\mathcal{P}_i| = \phi(\alpha_i, \epsilon)$ is dynamically scaled.[6] When $\mathcal{P}_i$ used to construct $\hat{Y}_i$ consists of mid-fidelity resources, we occasionally leverage high-fidelity resources with difference estimators to reduce the variance of the estimator.[7]

Where appropriate, low-fidelity resources are trained, fine-tuned, and tested using Y_i^*. They, in turn, generate valuations v_{ir} for $\mathcal{X}$. These valuations drive τ, inducing dynamics governed by $p_{t+1} = \mathcal{F}(p_t, \tau_t)$. After calibration via $g(\cdot)$, these valuations inform both π_{it} for the

[1] E.g. deep neural-networks , large language models, novice and expert human annotators, etc.

[2] Such characteristics can evolve over time. However, we assume they are relatively stable in the short-run.

[3] In reality noise is often heteroscedastic, arising from resource capabilities and latent task complexity.

[4] As reflected by $\text{tr}(\mathbf{N}_\kappa) \gg \text{tr}(\mathbf{N}_{\text{HF}})$

[5] For human annotators, v_{ir} is typically derived from structured ordinal responses $S_{ir\ell}$ over a template $\mathcal{Q}$.

[6] Task difficulty is frequently proxied by agreement (α_i) calculated over an initial rater subset.

[7] E.g. prediction-power-inference (PPI), double-sampling, etc.

production of Y_i^* and the deployment policy τ, which maps calibrated scores to exposure levels $a_\tau(x) \in [0, 1]$ interpreted as a multiplicative ranking factor. [8][9]

Low-fidelity resource performance is assessed using a self-normalized, importance-weighted confusion matrix $\mathbf{M}(\eta)$, yielding weighted precision $p_{\omega m}(\mathbf{M})$ and recall $r_{\omega m}(\mathbf{M})$ and other metrics. Evaluation is conducted under a counterfactual no-policy baseline by weighting observations using predicted counterfactual engagement $\widehat{E}_{it}$ to avoid demotion bias. When $\mathcal{P}_i$ used to construct Y_i^* consists of mid-fidelity resources, we refer to the reference label as noisy and denote it $\tilde{Y}_i$. To evaluate each panel, a subsample $\mathcal{D}_{n^{(2)}} \subset \mathcal{D}_{n,t}$ is adjudicated by high-fidelity resources. All resources are evaluated for fairness for protected partitions of entities, $\mathcal{M} \subset \mathcal{X}$ using the raw[10] false positive rate difference, $\Delta_{\mathcal{M},\mathcal{X}}\text{FPR}$, where the baseline represents the aggregate performance across the entire population $\mathcal{X}$. To optimize B_t, the system utilizes adaptive sampling schemes designed to maximize the informational utility of each acquired label.[11]

We can influence resource characteristics Φ_κ through treatments W [12] whose effects are often estimated via AB tests[13] that condition on additional features Z_i. Furthermore, we can identify resources r with unusually unsatisfactory characteristics relative to those of the same type κ and exclude them from $\mathcal{R}$.[14]

Uncertainty in all estimators $\hat{\theta}$ is quantified via a resampling operator $\mathcal{B}$, which induces bootstrap replicates $\mathcal{D}_{n,t}^{(b)} \sim \mathcal{B}(\mathcal{D}_{n,t})$, with estimates $\hat{\theta}^{(b)}$ summarized through their empirical distribution.[15]

Since $|\mathcal{X}|$ and most $|\mathcal{D}_{n,t}|$ are large, engines are configured to use Poisson sampling variants to mimic multinomial draws and facilitate high-throughput.[16]

Collectively, these components define a closed-loop system in which sampling, labeling, evaluation, and deployment interact through shared resource constraints and evolving population dynamics. The system objective is therefore to solve the constrained optimization problem $\min_{\tau,\mathcal{G}}\ \text{UCB}(\mathcal{L}(\tau, \mathcal{G}))$ subject to $\sum_{i \in \mathcal{D}_{n,t}} \sum_{r \in \mathcal{P}_i} c_{\rho(r)} \leq B_t$ and $\sum_{i \in \mathcal{D}_{n,t}} \mathbf{1}\{\rho(r) = \kappa,\ r \in \mathcal{P}_i\} \leq \Omega_\kappa, \forall \kappa$.

[8] $a_\tau(x) = 1$ corresponds to no demotion and $a_\tau(x) = 0$ corresponds to complete filtering.

[9] Where explainability is required, $g(v_{ir})$ may be accompanied by a vector of interpretive features.

[10] Non-engagement weighted

[11] E.g. multiple importance sampling (MIS), adaptive sampling, and sequential stopping rules.

[12] E.g. Changes to a template $\mathcal{Q}$, refinements to the labeling policy $\mathcal{G}$, etc.

[13] i.e randomized control trials

[14] When r corresponds to a human annotator, such interventions must comply with applicable co-employment laws.

[15] $\mathcal{B}$ is typically leverages a paired pigeonhole bootstrapping but can leverage other bootstraps.

[16] These are co-deployed with Gumbel-Max-style weighted reservoir sampling procedures.

2 Glossary

Population and Target

$\mathcal{X}$	Population of entities
x, x_i	Entity in the population
i	Entity index
t	Time index
Y_i	Latent true label (1 = positive class)
E_{it}	Engagement weight of entity x_i at time t
$p_t(x)$	Nominal (traffic-weighted) distribution, $p_t(x) \propto E_{it}$
$\mu_E = \mathbb{E}_{x \sim p_t}[Y(x)]$	Engagement-weighted prevalence

Sampling

X_i	Sampled entity
R_{it}	Sampling indicator
$\pi_{it} = \mathbb{P}(R_{it} = 1)$	Inclusion probability
$\mathcal{D}_{n,t}$	Sampled dataset $\{X_i \mid R_{it} = 1\}$
$q(x)$	Importance sampling distribution
$w_i = \frac{p_t(X_i)}{q(X_i)}$	Importance weight
$\mathcal{H}$	Set of strata
h	Stratum index
T	Time horizon for pooled sampling

Resources and Policies

$\mathcal{R}$	Set of resources
r	Resource index
$\mathcal{K}$	Set of resource types
κ	Resource type index
$\rho(r)$	Mapping from resource to type
$\Phi_\kappa = \langle c_\kappa, \Omega_\kappa, \mathbf{N}_\kappa \rangle$	Resource profile
c_κ	Marginal cost
Ω_κ	Processing capacity
$\mathbf{N}_\kappa$	Noise structure
B_t	Total budget
$\mathcal{G}$	Labeling policy

Annotation and Valuation

$\mathcal{P}_i \subset \mathcal{R}$	Panel of resources
v_{ir}	Valuation
$\mathcal{A}(\cdot)$	Aggregation function
$\hat{V}_i = \mathcal{A}(\{v_{ir}\})$	Aggregated valuation
η	Decision threshold
$\hat{Y}_i = \mathbb{I}(\hat{V}_i > \eta)$	Predicted label
$\tilde{Y}_i$	Noisy reference label
$\mathcal{D}_{n^{(2)}}$	High-fidelity subsample

Structured Human Annotation

$\mathcal{R}^{(H)}$	Human annotators
$\mathcal{Q}$	Annotation questions
ℓ	Question index
$S_{ir\ell}$	Ordinal response
v_{ir}	Derived valuation

Prediction and Deployment

$v(x)$	Scalable valuation
$g(\cdot)$	Calibration
$a_\tau(x) \in [0,1]$	Exposure multiplier
τ	Deployment policy
$p_{t+1} = \mathcal{F}(p_t, \tau_t)$	Population dynamics

Evaluation

$\mathbf{M}(\eta)$	Weighted confusion matrix
$p_\omega(\mathbf{M})$	Weighted precision
$r_\omega(\mathbf{M})$	Weighted recall
$\hat{E}_{it}$	Counterfactual engagement
$\mathcal{M}$	Protected subset
$\Delta_{\mathcal{M},\mathcal{X}}\mathrm{FPR}$	FPR difference

Objective and Optimization

$\mathcal{L}(\tau, \mathcal{G})$	Loss functional
$\mathrm{UCB}(\cdot)$	Upper confidence bound
$\min_{\tau,\mathcal{G}} \mathrm{UCB}(\mathcal{L})$	System objective
n_κ	Assignments per type

Uncertainty

$\hat{\theta}$	Estimator
$\mathcal{B}$	Bootstrap operator
b	Bootstrap index
$\mathcal{D}_{n,t}^{(b)}$	Resampled dataset
$\hat{\theta}^{(b)}$	Bootstrap estimate

Causal Variables

W	Treatment
Z_i	Covariates